

Amar Merabti

Data Scientist — Machine Learning & AI Engineer

merabtiamar91@gmail.com · merabtiamar.github.io · Paris, Île-de-France

Profile

Data Scientist seeking a permanent position (CDI), available immediately. M2 in AI & Data Science at Université Paris Cité, ranked 1st in my class (16/20). **One year of experience** as a machine learning developer at ENAGEO (Sonatrach group), then a final-year internship at Nricher on live product data: built an automatic quality-control chain that clears 68% of extracted values without human review, at 99% precision. Python, PyTorch, NLP/LLM, distributed Kafka/Spark pipelines.

Technical Skills

Programming: Python, SQL, Java (Swing), C/C++

Tooling: Git, Linux

Machine Learning: Scikit-Learn, linear and logistic regression, Ridge/Lasso, decision trees, Random Forest, Gradient Boosting (XGBoost, LightGBM), SVM, k-NN, Naive Bayes, MLP, K-means, DBSCAN, hierarchical clustering, PCA, t-SNE, Isolation Forest, calibration

Deep Learning: PyTorch, TensorFlow, Keras, Transformers, LSTM, U-Net, multi-GPU training, TabPFN, TabSTAR

NLP / LLM: BERT/DeBERTa, GPT-2, LLaMA, Mistral, DeepSeek, NLI, BGE-M3 embeddings, fine-tuning, RAG

Data Engineering: Kafka, Spark/PySpark, Pandas, NumPy

Databases: MongoDB, MySQL, SQLite, SQL/NoSQL

GPU compute: Tesla T4, P100, NVIDIA L4, batched inference, fp16

Viz & BI: Streamlit, Power BI, SSIS

Professional Experience

Data Scientist (final-year internship) — Nricher, Paris

Feb. – Aug. 2026

- Designed an automatic quality-control system for extracted product attributes (colour, dimensions, material, weight. . .) combining mDeBERTa-v3 textual inference with a TabPFN tabular model, across 4 client catalogues and 35 attributes: **68% of values cleared without human review at 99% precision**, cutting review volume by 42% on a loop that previously took 1 to 2 days per product category.
- Optimised GPU inference: batched forward passes, fp16 autocast, dynamic label resolution — **20 to 50× speed-up** on P100.
- Built a 30,919-row labelled dataset **with no manual annotation**, derived from 893,756 product/attribute pairs through an 8-stage reproducible pipeline (Python, Pandas).
- Benchmarked 3 architectures over 5,635 paired runs: **0.925 balanced accuracy** for the retained solution (TabPFN over NLI scores and PCA-reduced embeddings) against 0.898 for TabSTAR, ruled out despite its native text reading for a **×28 cost per run**. Calibrated probabilities and a decision threshold that can be moved after the fact without retraining: the operating point stays with the business.
- Wrote and ran rule- and lexicon-based attribute extraction algorithms, normalised values to the client reference schema, analysed the resulting check files.

AI Developer (Machine Learning) — ENAGEO (Sonatrach group), Boumerdès

July 2023 – July 2024

- Part-time fixed-term contract alongside the M1 SII: reconstructing well measurements (elasticity, density. . .) that are missing or costly to acquire, treating each depth-indexed series as a time series.
- Within-well imputation with LSTM and RNN models (trend, seasonality, cycles): **MAE 12.7, RMSE 15.2** (LSTM).
- Cross-well generalization (LSTM) vs. the internship: **correlation 0.52 → 0.82**, MAE 21.4 → 15.8, RMSE 27.6 → 20.2.

Machine Learning Intern — ENAGEO (Sonatrach group), Boumerdès

Feb. – June 2023

- Predicted geomechanical properties of reservoir rock from the well logs of 3 boreholes: linear regression / RBF-SVM / multilayer perceptron benchmarked with Scikit-Learn under cross-well generalization — **MAE cut from 27.4 to 21.4** by the MLP, correlation capped at 0.52 and judged too low for production.
- Delivered an application to the company's geophysicists: CSV import/export, model training, prediction of missing values and user account management — Java Swing + Python, MySQL, UML design.

Projects

- Real-time e-commerce recommendation engine** (M2 · 2025) — distributed streaming architecture Kafka → PySpark → MongoDB → Streamlit: 885K events, a catalog of 24,255 products, hybrid recommendation and online relevance scoring.
- Data quality pipeline** (M2 · 2025) — multi-criteria scoring (completeness, accuracy, consistency, uniqueness) contextualised by use case: regression-based imputation ($r \approx 0.78$), geometric reconstruction of GPS coordinates (Haversine formula), correction of temporal inconsistencies.
- Kabyle LLM trained from scratch** (personal project · 2026) — 36.3M-parameter decoder-only Transformer in PyTorch (Llama-style, 10 layers), custom BPE tokenizer, pretraining on 107M tokens on one Tesla T4, then fine-tuning.
- Open-source LLM benchmark** (M1 · 2025) — three 7B models (LLaMA 2, Mistral, DeepSeek-R1-Distill-Qwen) on GSM8K, HellaSwag and HumanEval via Ollama, on 4× NVIDIA L4: reproducible protocol and accuracy/latency trade-off analysis.
- Temporal expression extraction with Transformers** (M2 · 2025) — BERT token classification for temporal entity typing (TIMEX3 schema), evaluated across 3 corpora with contrasting registers (historical, clinical, tweets): fully automated end-to-end pipeline, F1 up to 95.5%.
- Time series imputation** (M1 · 2025) — 6 strategies compared on 3,127 electricity consumption series × 12,864 points (GPU STL, LSTM autoencoder, U-Net, Transformer): best MAE 0.262.

Education

2024–2026 **M1 & M2 Data, Knowledge and Intelligence (DCI), Ranked 1st in class** — Université Paris Cité

2020–2024 **B.Sc. Computer Science + M1 SII** — USTHB, Algiers

2020 **Baccalaureate, Experimental Sciences, with Honors** — Algeria

Languages

Languages: French (bilingual) · English (bilingual) · Arabic (fluent) · Kabyle (native)